

of reducing the need for data storage and enhancing the calculation speed, which were considered to provide a technical solution to a technical problem. In T.505/18 the board decided, in line with T.651/12, to recognise improved safety in the technical context of the invention, that of real-time route guidance of a driver of a vehicle, as being a technical effect, and the considerations regarding improved safety in a vehicle navigation system as being technical considerations made by the technical expert in navigation systems.

e) Artificial Intelligence and Machine Learning

In T.598/07 the invention concerned a heartbeat monitoring method, which was based on a neural network for the purpose of identifying irregular heartbeats. The board held that this made a technical contribution.

In T.1286/09 the invention related generally to the field of digital image processing and, in particular, to a method for improving image classification by training a semantic classifier with a set of exemplar colour images, which represented "recomposed versions" of an exemplar image, in order to increase the diversity of training exemplars. The board found that it involved an inventive step.

In T.1510/10 the invention concerned ranking information, particularly live web applications, based on interest and/or importance. The board had to consider whether using machine-learning algorithms could contribute to inventive step. The board highlighted that the claimed subject matter failed to define any particular method of machine learning – not even one was described in the application. Rather, machine learning was presented in the application as known. Thus, the Board decided that 'no inventive step can derive just from the use of machine learning. The Appeal was dismissed.

In T.1285/10 the invention related to a genetic analysis computing system with a method for diagnosing and recommending treatment for a physiological condition using artificial intelligence. The board held that it was common ground that use of artificial intelligence generally was already known. It was the use of hybridization information from an array of peptide nucleic acid probes, which was in question. The board did not come to a decision on inventive step, but observed that the claims were obvious in the light of the prior art.

In T.1784/06 the automatic classifying of abstract data records was held to be non-technical since the data records were classified for the non-technical purpose of billing. A valuable mathematical property of the algorithm could imply technical benefits but only when used for a technical purpose.

In T.755/18 the board held that if neither the output of a machine-learning computer program nor the output's accuracy contributed to a technical effect, an improvement of the machine achieved automatically through supervised learning to generate a more accurate output was not in itself a technical effect.

f) Linguistics and classification of texts

In T 1177/97 the board held that the use of a piece of information in a technical system, or its usability for this purpose, may convey a technical character to the information itself in that it reflects the properties of the technical system, for instance by being specifically formatted and/or processed. Such information when used in or processed by the technical system may be part of a technical solution to a technical problem and form the basis for a technical contribution of the invention to the prior art. Information and methods related to linguistics may thus in principle assume technical character if they are used in a computer system and form part of a technical problem solution.

Methods of text classification per se did not produce a relevant technical effect or provide a technical solution to any technical problem (T 233/09, T 1316/09, and T 1358/09).

In T 1358/09 claim 1 defined a method for classifying text documents essentially in terms of an abstract mathematical algorithm. The board noted that the classification of text documents was certainly useful, as it could help to locate text documents with a relevant cognitive content, but in the board's view it did not qualify as a technical purpose. Whether two text documents in respect of their textual content belong to the same "class" of documents was not a technical issue. The board noted that the same position was taken in decision T 1316/09, which held that methods of text classification per se did not produce a relevant technical effect or provide a technical solution to any technical problem (see also T 233/09). In T 1358/09 the board pointed out that not all efficiency aspects of an algorithm are by definition without relevance for the question of whether the algorithm provides a technical contribution. However, such technical considerations must go beyond merely finding a computer algorithm to carry out some procedure (see G 3/08, OJ 2011, 10; see also T 2418/12, T 22/12). In T 22/12 too, the board thus decided that the classification of emails as spam was not technical. In T 2363/16 the board cited T 22/12 and held that the classification criteria regarding which emails should be blocked were determined by the user of the system based on non-technical considerations regarding which emails the user did not want to receive.

In T 817/16 the board held that merely assigning a score to a document for a search engine in a memory-efficient manner was not a technical effect even if the score is somehow based on the frequency and the amount of changes made to the document. The appellant suggested that providing good scores improved the search results and therefore reduced the number of search queries, which amounted to a saving of resources. A similar argument was dealt with in decision T 306/10 in the context of recommendation engines. The board in that case considered that a reduction in the number of search queries and the corresponding saving of resources did not qualify as a technical effect of the (improved) recommendations, as they depended on subjective choices made by the user. It referred to decision T 1741/08, where the argument was made that a chain of effects could not be used as evidence of a technical effect if one of the links between the effects was not of a technical nature (but, for example, of a psychological nature).

g) Image processing

T 208/84 (OJ 1987, 14) was concerned with a method of digitally filtering a two-dimensional data array (representing a stored image). The board noted that a basic difference between a mathematical method and a technical process could be seen in the fact that a mathematical method or a mathematical algorithm was carried out on numbers (whatever those numbers represented) and provided a result also in numerical form, the mathematical method or algorithm being only an abstract concept prescribing how to operate on the numbers. No direct technical result was produced by the method as such. In contrast, if a mathematical method were used in a technical process, that process would be carried out by a technical means on a physical entity (which could be a material object but equally – as here – an image stored as an electrical signal) and would result in a certain change in that entity. The technical means might include computers comprising suitable hardware or appropriately programmed general-purpose computers (as similarly found in T 935/97).

In T 1286/09 the invention related to a method for improving image classification by training a semantic classifier with a set of exemplar colour images, which represented "recomposed versions" of an exemplar image, in order to increase the diversity of training exemplars. The board found that it involved inventive step.

In T 1455/16 the application related to retrieving images according to user specified arrangement information. The image data memory means was arranged to store position information of the image data, indicating the shooting spot of the image and recommending a shooting spot at which such an image could be captured. The board held that the text of the application did not reflect the application of the position information in a technical task of guiding the user to a point in space. Therefore, it could not be said that in the present case providing position information to a user was based on technical considerations. The mere fact that the position information concerned physical elements in the real world was not sufficient for establishing a technical contribution of providing the position information (T 154/04, OJ 2008, 46; T 2035/11; T 670/16; different in T 572/15).

h) Security

In T 1326/06 the board held that methods of encrypting/decrypting or signing electronic communications had to be regarded as technical methods, even if they were essentially based on mathematical methods. It referred to T 953/04, where the board had stated that the use of cryptographic methods in the technical context of electronic data processing and communication certainly had technical character. The board in T 1326/06 also referred to T 27/97, where the board had stated that a method of encrypting or decrypting a message represented in the form of a digital word using RSA-type public-key algorithms was not excluded from patentability under Art. 52(2) and (3) EPC, even if based on an abstract algorithm or mathematical method.

In T 2101/12 the application related to a computer implemented invention concerning a method and system for providing an electronic signature and authentication. The board considered it common general knowledge that documents, such as a will or a contract

between parties, may be signed at a notary's office. With this common general knowledge process as starting point, the board held that the skilled person would, as a result of a straightforward automation of this process, arrive at the subject-matter of the invention at hand without any inventive activity. The board further considered that the process as described contained sufficient detail to allow the skilled person to arrive at the claimed subject-matter using otherwise nothing more than conventional automation techniques.

In T.1461/12 the application related to a system and method for auditing software usage. Claim 1 specified that running of the copy of the soft-ware would be terminated after expiry of the limited unlock key. The board found that the differences to the prior art related primarily to the terms and conditions of the license in question. In the board's view, it was up to the rights owners whether they wanted to limit the number of "copies" made of a given piece of software and to grant one set of rights for the first number of copies and a different, limited set of rights subsequently for a second number of copies. Moreover, the board took the view that the rights owner would make that choice according to legal, economic and administrative considerations rather than technical ones. The board therefore considered that the content of the license did not contribute to the technical character of the invention.

The board in T.548/13 held that the only distinguishing feature, requiring the security features to show different views of the same image on the front and back of a value document, was non-technical. It conceded that it made the document more difficult to counterfeit, but two different motifs could just as easily be used.

In T.1749/14 the invention was in the field of mobile point-of-sale (POS) terminals for carrying out transactions, e.g. involving a credit card. The invention tried to avoid the customer's sensitive information becoming known if a merchant's device was tampered with by allowing a transaction to be carried out without the customer having to present account information and the PIN to the merchant. The board held that the invention required a new infrastructure, new devices and a new protocol involving technical considerations linked to modified devices and their capabilities as well as security relevant modifications of the transfer of sensitive information using new possibilities achieved by the modifications to the previously known mobile POS infrastructure. The board considered the security relevance of the modifications to contribute to the technical character of the present invention.

9.2.12 Discoveries

In T.1538/05 the applicant claimed to have discovered a magnetic force that was hitherto unknown, and as a consequence thereof, established a number of theories. The board held that the discovery of a new magnetic force was not patentable.

In T.2677/16 the board held that identifying a drug discovery target was not technical. It held that a drug target was not a therapy, but merely a promising direction for future research. The discovery of a drug target may well turn out to be a valuable scientific discovery, but neither discoveries nor science have technical character as such, as is

explicitly enshrined in Art. 52(2)(a) EPC. This should not be confused with the invention of a drug, which would represent a technical purpose.

In T 1789/13 the board held that the "weather" was not a technical system that the skilled person could improve, or even simulate with the purpose of trying to improve it. It was a physical system that could be modelled in the sense of showing how it worked. This kind of modelling was held to be rather a discovery or a scientific theory, which are excluded under Art. 52(2)(a) EPC and thus do not contribute to the technical character of the invention.

9.2.13 Human Perception

In T 619/12 the board held that the perceptual processes taking place in the mind of a test person presented with odours in an odour selection test did not constitute mental acts within the meaning of Art. 52(2)(c) EPC. Nonetheless, human perception phenomena could not be qualified as being of a technical nature.

In T 339/13 the application was concerned with a virtual pet that was capable of giving "haptic feedback". Noting that the owner of a toy had to be willing to accept the toy's behaviour as real, the board accepted that achieving a reliable and reproducible perception of a physical interaction was a technical problem and found that the invention solved this problem by technical means, namely a reciprocating cursor movement and haptic feedback.

9.2.14 Games

In T 1023/06 the invention related to a method of operating an electronic video poker machine. Following the approach of T 49/04 the board held that an improvement in readability, which related to how "cognitive content" was presented, constituted a technical contribution.

In T 717/05, T 42/10 and T 1281/10 the boards had to examine whether the amusement of the player could be a technical problem. In T 717/05 the board held that amusement was the psychological purpose of a gaming apparatus and thus was the relevant objective technical problem to the extent that the enhanced amusement was achieved by technical features of the claim. In T 42/10 and T 1281/10 the boards denied this.

In T 336/07 the board held that the mere fact that subject-matter (here rules for playing games), which is excluded per se under Art. 52(2)(c) EPC 1973, is technically implemented could not form the basis for inventive step (see also in this chapter I.D.9.2.3). Inventive step can be based only on the particular manner of implementation of such subject matter. To this end it is therefore necessary to ask how the per se excluded subject matter was implemented. A consideration of the particular manner of implementation has to focus on **any further technical advantages** or effects associated with the specific features of implementation over and above the effects and advantages inherent in the excluded subject-matter (see also T 1543/06). In T 337/07 the board concluded that the subject-matter of the claims did not involve an inventive step. In T 1782/09 where the